Learning Incomes

- Familiarity with the limit definition of the derivative.

Note: Throughout this tutorial we deal with various limits and derivatives. Feel welcome to use any results and techniques from your previous studies on these concepts.

1. **A Differentiable Piecewise Function**

   Let $k \in \mathbb{R}$, and let $f : \mathbb{R} \to \mathbb{R}$ be defined by

   $$f(x) = \begin{cases} kx - k + 1 & x < 1 \\ x^3 & x \geq 1 \end{cases}$$

   Using the limit definition of the derivative and Theorem 5.7, determine all values of k for which f differentiable at $c = 1$.

2. **A Flawed Proof**

   Let $f : \mathbb{R} \to \mathbb{R}$ be defined by

   $$f(x) = \begin{cases} 4 + 2x & x \leq 0 \\ x^2 + 2x & x > 0 \end{cases}$$

   Claim. f is differentiable at $c = 0$.

   Proof. We show $f'(0) = 2$ and thus f is differentiable at $c = 0$.

   To show

   $$\lim_{x \to 0} \frac{f(x) - f(0)}{x} = 2$$

   we examine the left and right side limits.

   When $x < 0$ we have $f'(x) = 2$. Therefore

   $$\lim_{x \to 0^-} \frac{f(x) - f(0)}{x} = 2.$$

   When $x > 0$ we have $f'(x) = 2x + 2$. Therefore $f'(0) = 2$. Therefore

   $$\lim_{x \to 0^+} \frac{f(x) - f(0)}{x} = 2.$$

   Since

   $$\lim_{x \to 0^-} \frac{f(x) - f(0)}{x} = \lim_{x \to 0^+} \frac{f(x) - f(0)}{x} = 2$$

   we have

   $$\lim_{x \to 0} \frac{f(x) - f(0)}{x} = 2.$$

   Therefore $f'(0) = 2$. Thus f is differentiable at $c = 2$. ■

   Draw a picture of the function and conclude that the claim is false. Find the error in the proof.

3. **Derivatives of Increasing Functions**

   Let $f : [a, b] \to \mathbb{R}$ be differentiable on $[a, b]$. Using the limit definition of the derivative, prove that if f is monotone increasing on $[a, b]$, then $f'(c) \geq 0$ for all $c \in [a, b]$.

   (f is <u>monotone increasing</u> when for all $x, z \in [a, b]$ such that $x \leq z$ we have $f(x) \leq f(z)$.)

4. **Locally Non-Constant** Let f be differentiable on $[a, b]$ and let $c \in (a, b)$.

   (a) Prove that if $f'(c) \neq 0$, then there exists $\delta > 0$ so that $f(x) \neq f(c)$ for all $x \in (c - \delta, c + \delta)$ with $x \neq c$.

   (b) Is the converse of the statement in (a) true?

5. **A Discontinuous Derivative**

   Consider the function $f : \mathbb{R} \to \mathbb{R}$ given by

$$f(x) = \begin{cases} x^2 \sin \dfrac{1}{x} & x \neq 0 \\ 0 & x = 0 \end{cases}$$

   (a) Determine if f is continuous on $\mathbb{R}$?

   (b) Find the derivative at all points $c \neq 0$.

   (c) Using the limit definition of the derivative, find the derivative at $c = 0$. Appeal to left and right side limits. Use any limit techniques you know from this course and from Calculus 2.

   (d) Determine if $f'(x)$ is continuous at 0.